

The Virtual Precinct: Place-Anchored Retail, Encounter-Based Demand, and Ground Rent as Platform Value Capture — A CamboVerse Case

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No sales, revenue, conversion, traffic or adoption figures appear in this paper, because none exist. Screenshots of a running prototype show what has been built, and nothing else.

Abstract

Three-dimensional shops you can walk around have been announced as the future of retail twice — in virtual worlds around 2008, and in the metaverse around 2022 — and abandoned twice. We do not make the announcement a third time. We argue that both waves failed at the same point: they built places to look, not places to buy, and so never met the real questions a shop like this raises.

This paper asks those questions using CamboVerse, an open-source Virtual Cambodia — a metaverse project for the whole country, in which heritage sites are one part — with a working public prototype. Our case is the market at one of its heritage sites. When a shop sits inside a virtual version of a real place, and you reach it by visiting that place, you find products by walking past them rather than by searching. We argue that this is a property of the shop's design, not of the shopper's mood, and that it changes what people are exposed to. When the platform then routes your order to a shop in your own country, takes no payment, and charges no referral fee, three things follow. The seller is not known until checkout and has no contract with the platform. The platform earns nothing from the sale — unlike affiliate links or search ads, which charge for referrals. And the sale is recorded as ordinary domestic shopping in the buyer's country,

invisible to the statistics that measure digital trade.

We then ask what such a platform could charge for, and argue that the only remaining answer is the ground: rent for stall space in a virtual market, priced on the visitors the platform's own content brings. This would keep the no-payment rule intact. It is also only a proposal, because the platform's design currently defines a router and a list of providers, and nothing that could be rented. It would inherit the problem that emptied the 2008 virtual worlds, where rent rose until merchants left. And in Cambodia it would rest on a claim that can be collected but not enforced in court.

We state each claim as either established by the design, borrowed from earlier research, following logically from stated premises, or a guess to be tested — and we keep these apart. The paper makes no empirical claim. Nothing has been sold.

1. Introduction

Imagine a visitor in Ohio. She opens a link on her phone and finds herself walking along the causeway of a famous Khmer temple. The scene runs on whatever device she has. It works on a cheap handset, and it scales up to full detail, or to a VR headset, if she owns one. At the temple gate there is a market. She wanders in, looks at a few stalls, and buys a bag of coffee. A shop in her own town delivers it. The screen tells her so before she pays.

Almost nothing about that purchase is normal. She did not search for coffee. She was not looking for a brand. No algorithm recommended anything. She bought something because she walked past it, in a place she was visiting for another reason — which is how shopping worked for thousands of years before it became a search box. The shop that filled her order was not chosen until she reached the checkout, and it has no contract with the platform that brought her there. And the platform did not take her money. It pointed her to a local shop and stepped aside.

The easy reading is that this is online shopping with nicer graphics. We think that reading is wrong. But we start by admitting how much reason there is to believe it.

1.1 This idea has failed twice before

The claim that a 3D shop you can walk around is a new kind of retail is not new, and its history is not encouraging.

Bourlakis, Papagiannidis and Li (2009) argued that retail was moving through three spaces — physical shops, websites, and virtual worlds — and that retailers could operate in all three intertwined spaces at once. Research at the time backed them. Animesh and colleagues (2011) surveyed Second Life residents and found that the spatial design of the world raised their intention to buy virtual goods. Haenlein and Kaplan (2009) found that visiting a brand’s shop in a virtual world improved how people felt about the brand, and their intention to buy from it, in real life.

Then the shops closed. Kuntze and colleagues (2013) looked at the top hundred retailers and found only one still had a shop in Second Life. Toyota, Reebok, American Apparel and Adidas had all left by 2008 — while the studies above were still being published. Among the reasons they give is one this paper must answer for: the rent on virtual land kept rising until it wiped out the point of being there.

It happened again. The metaverse wave of 2021–2023 produced serious research agendas (Yoo and colleagues 2023), and then retailers pulled out again. A 2025 survey of VR shopping apps (Elsholz and colleagues 2025) found most are still “predominantly showrooms without purchase possibilities.”

That phrase is our whole answer, and we put it here so the reader has it early. Both waves built places to look. Neither built places to buy. A showroom cannot tell you how a shop really works — who sells, who gets paid, who takes a cut — because none of that happens until money moves. What changed is that in May 2025 a large gaming platform — one with tens of millions of daily users — opened real shopping inside its 3D worlds, with well-known retailers selling physical goods there. Whatever one thinks of that platform, the fact is new. The questions this paper asks could not have been asked in 2009 or 2022, because nobody was selling anything.

1.2 What we claim

We make four claims. For each, we say what is already known.

First: people find things by walking past them. In a shop anchored to a real place, you discover products because you are standing near them, not because you searched or wanted a brand. Researchers know that shoppers sometimes search and sometimes browse (Moe 2003). They know that changing how people find things changes what sells (Brynjolfsson, Hu and Simester 2011). What nobody has said is that “finding by walking past” can be a property of the *shop itself* — built into its design, not a mood the shopper is in.

Second: the place is real. A shop attached to a real location that people visit for its own sake is different from a brand’s shop in a made-up world, and different from a

3D copy of an existing store. All three exist. Nobody has separated them or asked whether they work differently.

Third: a platform that refuses to take a cut. Sending a customer to a shop you have no contract with is ordinary. Affiliate links, search ads and map listings do it every day — and each charges a fee for the referral. This platform does not. It routes the order to an unaffiliated local shop, takes no payment, and charges no fee. So it captures nothing from the sale at all. We do not call this a theoretical failure. It is a choice, made twice over by two design rules, and the rest of the paper works out what follows from it.

Fourth: what is left to charge for is the ground. If the platform cannot take a share of sales, the only thing it controls is space in the market. We work out what charging rent for that space would mean — and we are careful to say that the platform’s current design does not yet contain anything that could be rented.

Section 2 reviews what earlier research already covers. Section 3 explains our method. Section 4 describes the platform. Sections 5 to 7 build the argument. Section 8 discusses what it means, Section 9 says what we do not know, Section 10 says what should happen next, and Section 11 concludes.

2. What Is Already Known

We owe the reader a clear account of what other researchers have already said, because much of what looks new here is not.

2.1 Shops in three-dimensional space

It is tempting to say this paper breaks the old textbook split between “store” and “non-store” retailing. We resist that, for two reasons. The split was never a real theory — the most recent review of retail formats (Gauri and colleagues 2021) organises formats by whether they reduce effort or enrich experience, not by whether they have a building. And official statistics retired the category in 2022, merging online and physical retail rather than adding a third kind. A category that disappears by merging is not evidence for a new third category.

What earlier work does own is the observation that space still matters when you sell in 3D. Bourlakis and colleagues (2009) said virtual worlds were a third retail space. Gadalla, Keeling and Abosag (2013) showed that service quality in a 3D shop has dimensions a website does not have. We build on these. We do not restate them as discoveries.

2.2 A shop is a place, even on a screen

For over thirty years, marketing research has treated the shop as a place with an atmosphere that affects how people behave — first the physical shop (Bitner 1992), then the website (Harris and Goode 2010). This research is the biggest obstacle to our argument, and the obstacle is subtle. Its natural view is that a 3D shop is simply a richer website: more immersive, further along the same scale. On that view

our claim is that “more immersive is better,” which is neither surprising nor, given the history in Section 1.1, convincing.

Our answer cannot be that our shop is more immersive. It has to be that something changes in kind. We find that change not in the atmosphere of the shop but in *how a product ends up in front of you at all* — something the atmosphere research takes for granted.

2.3 How people find things

Three lines of research come close to this question and stop short.

Consumer research distinguishes shoppers who are searching for something from shoppers who are just browsing (Moe 2003). But that is a property of the shopper, not of the shop; the same website serves both.

Economics gives the nearest real precedent. Brynjolfsson, Hu and Simester (2011) showed that when searching gets cheaper, sales spread out toward obscure products — the way people find things changes what sells. But their model treats online shopping as cheaper searching. It does not describe a shop where searching is not how you find things.

So the gap is specific. Nobody has treated “finding by presence” as a built-in property of a retail format with its own effect on what people end up seeing. That is where Section 6 goes.

2.4 Virtual-world shops and why they closed

Section 1.1 told the story. Two points about how to read it.

First, the findings from that period were not proven wrong. They were abandoned. Animesh and colleagues’ result about spatial design, Haenlein and Kaplan’s result about brand attitude, and Hassouneh and Brengman’s (2015) typology of virtual store atmospheres were never overturned. The platforms that hosted them emptied out. A field that mistakes a dead platform for a dead theory throws away good results, so we treat these as standing findings.

Second, Kuntze and colleagues name rising virtual land rent as one cause of the exit. Section 7 proposes ground rent as the way this kind of platform could earn money. So we are proposing, as the solution, something documented as a cause of the last collapse. We do not hide from that. It is why Proposition 8 exists.

2.5 Shops that exist because a place draws visitors

In the physical world, shops that exist only because tourists come to a place are a known type. Murphy and colleagues (2011) call them tourist shopping villages: places whose retail appeal rests on visitor flow rather than on everyday resident shopping. The market at a temple gate is one. What we describe is its virtual version, and this is the right comparison.

One study in this area cuts directly against us, and we must face it rather than cite it politely. Su, Lai and Huang (2024) compared an offline-only channel with a dual channel (offline plus the Internet). Each works through a different

path — the dual channel through convenience and usefulness, the offline channel through scarcity and value — and authenticity decides which path operates. When authenticity is high, only the offline path does. Read simply, that says the goods most tied to a place resist online sale — which would end a paper about selling place-tied goods online.

Their finding is more specific than that summary, and we do not lean on the summary. Their design compared two channels: offline only, and offline plus the Internet. A shop rendered inside a virtual version of the place was not one of them. So their result does not speak to the channel we describe. It tells us that adding a generic online option does not carry high-authenticity goods; it does not tell us what a place-rendered shop would do. That is a narrower claim than saying their result identifies a gap our design fills, and it is the one the published abstract supports. Whether their Internet condition carried any cues of place — which would bear on how far the two channels differ — we could not determine from the abstract, and we were unable to obtain the full text. We say so in Section 9.

2.6 Rent and location

The oldest idea we use is the simplest. Land near a busy place is worth more than land far from it, because more people pass by. The owner of the busy place can, in principle, charge for that.

Does this hold in a virtual world where people can teleport? Goldberg, Kugler and Schär (2024) found that it does, with a twist that matters for Section 7. Location premiums survive, but not because of travel cost — because of how many visitors a spot is expected to attract. And there is a *teleportation threshold*: beyond a certain distance, people teleport instead of walking, and distance stops mattering. Location matters as a proxy for attention, not as a walk.

What nobody has studied is a market anchored to a real heritage site, entered only by teleport, so that every visitor arrives at the same spot. That is the case in Section 7.

3. Method

3.1 What kind of paper this is

This is a conceptual paper. In Jaakkola’s (2020) terms it is a *theory adaptation*: it takes ideas built for other purposes — retail format theory, the economics of how people find things, research on new ways of entering foreign markets, and the theory of location rent — and applies them to a kind of transaction those ideas were not built for. Following Cornelissen (2017), we deliver the argument through a typology (Section 5) and a set of numbered propositions (Sections 6 and 7). MacInnis (2011) explains why this kind of contribution is worth making.

3.2 How we use the case

Our example is **CamboVerse**, an open-source metaverse project for Cambodia — a Virtual Cambodia. Its scope is the country, not a single theme: heritage sites, cultural modules

such as craft, dress, farming, ceremony and language, learning spaces and markets are all parts of one platform. We use the market attached to one of its heritage sites as the illustrative case, because that is where the commerce layer is built. Its architecture is published in full, and a working prototype is open to anyone at `preview.camboverse.world`. We use it as an *illustration*, not as a test. We chose it for two reasons. Every idea in our argument corresponds to a part of the published design that a reader can inspect. And because the prototype runs, the arrangement we describe can be seen rather than only read about — which matters for an argument about the *shape* of a shop.

The commerce part of the platform is at proof-of-concept stage. No one has bought anything. We make no claims about sales, conversion, revenue or adoption anywhere in this paper, and a reader who needs such evidence will not find it here.

3.3 What we checked before claiming anything

We searched five bodies of research before settling on what to claim: cross-border e-commerce, metaverse and immersive retail, international market entry, local production, and virtual-world retail. That search showed that several things an earlier version of this argument claimed were already known. In particular, the idea that a brand or design can cross a border while the physical product is made locally is the founding idea of franchising theory, is written into international trade law, and has its own literature in innovation studies. We concede these in Section 2 and Section 8 rather than let a reader discover them.

3.4 Four kinds of claim

Every substantive statement in this paper is one of four kinds, and we keep them apart.

A *specified* claim is true of the published design and can be checked against it. A *carried* claim is established in earlier research and imported. An *analytic* claim follows logically from stated premises — it is not an empirical finding, but given the premises it holds. A *conjectural* claim is a reasoned guess about how people will behave that nothing here establishes; these are offered to be tested.

The line between analytic and conjectural is the one most easily blurred to our advantage, so we state it sharply. That a platform which has no contract with sellers and takes no payment captures nothing from the sale is *analytic* — it follows from the two rules. That buyers will find a locally made product less authentic is *conjectural* — it is a claim about people, and we have not asked any.

3.5 Who we are

We are not independent of the case. One author directs the CamboVerse Center, an institute at the National University of Management, and two of the four are the engineers who built the system in Section 4. We state these as affiliations, not as claims about who governs the project.

This cuts both ways, and a reader should weigh both. It strengthens Section 4: the people who built the system describe it, and every statement there can also be checked against the public code and the public prototype. It weakens Sections 7 and 8: we have an interest in the arrangement looking important, and we are not neutral observers of something we made. The conjectural propositions are where that interest could most distort our judgement, and they are exactly the ones we have not tested. A reader who discounts them on those grounds is reasoning correctly.

4. The Platform, Described

This section describes. It does not argue.

4.1 The design rules

CamboVerse has some unusual fixed rules. The minimum device is a low-cost Android phone on a 4G connection, in a browser — stated as a pass/fail requirement, not a hope. Nothing is installed: an experience opens from a shared link. Only open standards are used.

It is important not to mistake this floor for a ceiling. The cheap phone is a minimum, not a description of the experience. The design defines three view modes: a reduced mode that must run on the minimum device, a full-detail mode for capable hardware, and an immersive VR mode over WebXR, with photorealistic capture as a stated direction. The rule is that lower modes reduce detail but never content, so the same scene, the same points of interest and the same market exist in all three.

For our argument the consequence is not that the shop looks poor. It is that whatever the shop does, it cannot rely on looking good, because it has to work on the cheap phone too. It has to work through *layout* rather than *polish*. That suits a paper arguing that the shape of the shop, not its atmosphere, is what matters — and it makes the argument stronger, because it has to hold at every level of quality.

4.2 How an order is routed

Commerce is exposed as a routing service, not a marketplace. An order request carries a type — delivery, purchase, booking or ticket — plus the buyer's country and an optional buyer identifier. The response names a provider in the buyer's region, the payment method used there, the delivery service used there, and a link to hand the buyer over. A separate service lists the providers by type and region.

Two details matter. The country is detected at the network edge, at country level only, and is shown to the buyer on screen. And every response carries a fixed marker: settlement happens outside the core, never inside it.

4.3 The rule against taking payment

The design states as a governing rule that no part of the core takes a payment. The core defines the request and response and runs a reference router; payment happens through outside providers. The core keeps no record of

orders — applications built on top keep their own.

We flag this because Section 7 turns on it. This is not an unbuilt feature. It is a rule the design imposes on itself, and the capture problem in Section 7.1 follows directly from it.

4.4 Where shops may go

The design confines commerce to market zones next to a heritage site and forbids it on sacred ground. This is a land-use rule inside a virtual world, with a real-world reason: real temples have markets at their gates, not in their sanctums. We note it because Section 7 proposes charging rent for space, and the space is bounded by a rule that is cultural, not commercial.

4.5 What we saw in the prototype

A public prototype runs in a mobile browser. What follows is what we *observed*, not what we read in the design, and the figures record it so a reader can repeat the observation.

You start at a map of Cambodia with pins for heritage sites, some marked live, alongside cultural modules — craft, dress, farming, village life, ceremony, language (Figure 1). Tapping a live site opens a 3D scene with labelled points of interest and a row of equal options: a guided tour, a view of the site in an earlier century, a walk-in mode, an AI guide, a photorealism toggle, a VR mode, and a market (Figure 2). The market is one option among several, no more prominent than the rest, and the visitor arrived at the site for reasons that need have nothing to do with it.

Tapping the market teleports you through a gateway into a round plaza with a fountain in the middle and kiosks around the edge (Figure 3). The merchants carry invented names. There is a cart.

Three things here bear on the argument. The market is reached from the site and is separate from the temple — the land-use rule of Section 4.4 made visible. The kiosks sit in different positions around a plaza, so which one you see depends on where you stand and which way you face — the mechanism of Proposition 1a. And all of it opens in a browser from a link, in reduced mode on a cheap phone and in richer modes where the hardware allows.

4.6 The checkout, observed

Tapping a kiosk opens a product list with prices and an add button (Figure 4). Items go into a cart. The cart shows a control labelled FULFILLED FROM naming a country, and under it the payment method and delivery service for that country. Change the country and both change (Figure 5): one choice shows a domestic instant-payment system and local delivery firms; another shows card payment through an international processor and that region's delivery apps. Seven countries are offered. The order button says “demo.”

This is the whole idea in one control. Same goods, same prices, same merchant; but the country of fulfilment, the payment system and the delivery firm all change together, and the visitor is told before ordering rather than after.

We show it so a reader can see the mechanism rather than take our word. It does not show the system *working*: the button says demo, and a demonstration of a routing rule is not the rule in operation. What the figure shows is that the idea has been specified precisely enough to build and display — more than a description, much less than evidence.

Two things about this screen matter later.

The total does not change when you change the country. Fine in a demo with a mock catalogue. Impossible in a real deployment, and who fixes it matters. If the platform sets one price for every region, it is setting prices — a marketplace function that moves it toward the very position Section 7.1 says it has declined. If the local merchant sets the price, the cart cannot show a total until the region is known. The current screen does both — a region chooser and a fixed total — which is less an open question than proof that the commerce object is still a mock. We record it as such.

The country is chosen, not detected. The design detects it; the prototype offers a menu. Sensible for a demo. But a buyer who can choose the fulfilment country may choose on price or speed rather than on where they live, which turns a routing rule into a marketplace of countries.

4.7 What the prototype does not show

The order is labelled a demo. We did not see a completed purchase, a payment, or a delivery. Whether the provider list behind the country menu points to real shops, and whether the handover works, we did not see and do not claim.

No transaction has happened. Nothing has been sold. No number for traffic, engagement, conversion or revenue appears anywhere in this paper. A running interface is evidence of what has been built, and of nothing else.

5. The Building Blocks

5.1 Three places that usually coincide

Normal retail does not need to separate three things, because they happen in the same place. Here they come apart.

The *place of demand* is where the wish to buy forms — here, a virtual version of a Khmer site. The *place of fulfilment* is where the physical product comes from — here, the buyer's own country. The *place of settlement* is where the money is taken — here, an outside payment provider that the platform has deliberately kept outside itself.

In a normal shop all three coincide. In ordinary cross-border online shopping the first two coincide abroad. Here, demand forms in one country's virtual representation, the product ships within the buyer's country, and the money is taken by neither the origin nor the platform.

5.2 Three kinds of virtual shop

Earlier work has studied three arrangements without separating them.

A *brand-world shop* is a flagship in a made-up world; you go there for the brand or the world. A *store twin* is a



Figure 1: Entry: a map of heritage sites alongside cultural modules. Live sites are marked. The visitor's first act is choosing a place, not a product.

3D copy of an existing shop; you go there for that shop. A *place-anchored market* is attached to a virtual version of a real place that exists for reasons other than shopping; you go there for the *place* and come across the market.

The third differs in what brings the visitor. In the first, brand loyalty does. In the second, an existing relationship with the shop does. In the third, the attraction does — and the attraction is owned by the platform, not by any merchant. That asymmetry is the origin of both the demand mechanism in Section 6 and the capture mechanism in Section 7.

5.3 Who takes a cut, and how

The typology that does the theoretical work asks, for each way of making something locally that originated elsewhere: how does the originator get paid?

Arrangement	How the originator is paid
Franchising, licensing, direct investment	Royalty, fee, equity, supply contract — built in
Design-global-manufacture-local	Nothing, <i>by choice</i> : shared designs, deliberately not for profit
Transaction-capturing spatial platform (Roblox, from 2025)	Contract with the merchant, platform takes the payment, keeps a share
Settlement-abstaining market (this case)	Nothing, by design : no merchant contract, no payment taken, no referral fee

The third row is a general-purpose gaming platform that sells physical goods inside user-built worlds, contracts with merchants, and takes a share of each sale; Roblox's commerce launch in May 2025 is the live example. It proves spatial retail can sell at scale, and it is the opposite design:



Figure 2: A heritage site with labelled points of interest. The market is one option among several — tour, past view, walk-in, AI guide — and no more prominent than the rest.

it takes its cut exactly where this case declines to. That contrast is the useful one. The fourth row is not the only way spatial retail can exist. It is how it exists when a platform refuses the take-rate.

The gap between the second and fourth rows is the paper's theoretical opening. A design-commons model (Kostakis and colleagues 2015) gives up profit as a principle, and that is coherent. This case gives it up as the accidental result of two separate design decisions — routing by geography, and not taking payment — neither of which was made in order to give anything up.

5.4 Three roles, and which ones the design actually defines

The propositions in Sections 6 and 7 involve three roles, which must be kept apart. The *originator* owns the attraction and brings the visitor. The *tenant* occupies a stall and would pay rent for it. The *fulfiller* supplies and delivers the physical product in the buyer's country.

In a normal shop the tenant and the fulfiller are the same business. Here they need not be, and the combinations differ. If the tenant is a brand and the fulfiller is that brand's regional operation, rent makes sense, and the fulfiller has no contract *with the platform* while still being tied to the tenant. If the tenant is itself the local fulfiller, routing to a different country cannot pick a new merchant without emptying the stall. If there is no tenant, only a list of providers, there is nothing to rent.

What the design defines today. The published architecture defines a router and a provider list. It defines no stall,



Figure 3: The market plaza, entered through a gate from the site. Kiosks sit in different positions around a round plaza; merchant names are placeholders. What you see depends on where you stand — Proposition 1a.

no tenancy, no lease, and no market-zone allocation. The rent proposal in Section 7 therefore sits on top of a design that does not yet contain the thing being rented, and would need a second kind of contract the design does not have. We say so here so that Section 7 cannot imply otherwise, and Propositions 6 and 9 are labelled accordingly.

5.5 Definition

We define a *place-anchored virtual market* as a shopping area inside a virtual version of a real place, in which what the visitor sees is determined by being there rather than by searching or seeking a brand; in which the physical product is supplied by a provider in the buyer’s own country; and in which the platform neither has a contract with that provider nor takes the payment.

5.6 What we assume

The following are assumed, not claimed. Each is established elsewhere and none is our contribution. There is a virtual place that conveys a real place’s identity. Buyers are spread across countries. The product has substitutable suppliers in each region. Being present in an immersive scene raises willingness to buy, as tourism research has found (Tussyadiah and colleagues 2018). Local supply is cheaper than cross-border shipping, as trade economics establishes. Outside these conditions the model makes no claim.

One assumption deserves a note here, and it is about who benefits rather than whether the mechanism exists. The research on presence was mostly done with headsets and

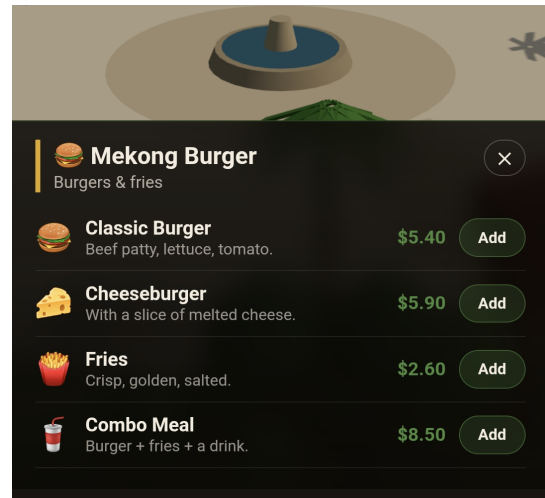


Figure 4: A kiosk product list. The goods are placeholders used to test the routing, not a sign of what will be sold; Section 6.2 explains why their lack of any link to the place is nonetheless useful for one test.

powerful computers. That maps onto the full-detail and VR modes of Section 4.1, where the assumption stands on firm ground. What is untested is the reduced mode — and the visitors who can only reach that mode are exactly the people a public-good project exists to serve. We therefore expect presence, and with it the strength of the demand mechanism, to *vary by mode*. That is a measurable variable, not a hidden assumption, and it is measurable on a platform that already ships three modes. If the mechanism works in VR and fails on the cheap phone, the format works but its benefits go to the better-equipped — a finding a public-good platform would want to know.

6. Demand: Finding Things by Being There

Proposition 1a (analytic). *In a market with no search box and no recommendations, which stall you see is determined by where you stand and which way you face, not by what you came wanting. So what visitors are exposed to does not depend on what they already wanted.*

Proposition 1b (conjectural). *Demand generated this way is more concentrated on a few stalls than demand generated by searching the same catalogue.*

We separate these because they are different kinds of claim. 1a follows from the interface: remove the search box and the recommender, and position is what is left to decide what you see. 1b is a claim about people. The design can put a stall in front of you. It cannot make you want what is in it. People walk past market stalls all day without buying.

Under search, you see what you asked for. Under recommendation, you see what a model predicts you want. In both, exposure is shaped by prior demand, stated or inferred. Under encounter, exposure is shaped by *position*. A stall near the entrance is seen by everyone who enters, whatever they

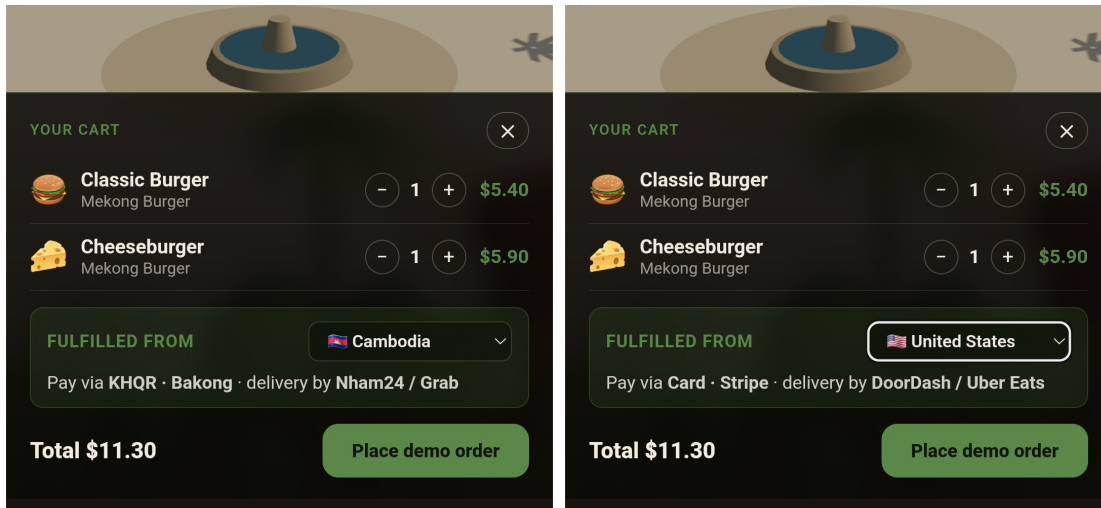


Figure 5: The idea in one control. Same cart, same merchant, same total; changing FULFILLED FROM changes the country of fulfilment, and with it the payment system and the delivery firm. Left: a domestic instant-payment system with local delivery. Right: card payment through an international processor with that region’s delivery apps. The button says demo.

wanted. This is the ordinary property of physical markets, and it is what searchable commerce removed.

The consequence runs opposite to the best-known result in this area. Brynjolfsson and colleagues (2011) showed that cheaper searching spreads sales toward obscure products, because people can find things they already want. Encounter has no such mechanism. It concentrates attention on well-placed stalls regardless of what anyone wants. So we expect market demand to be *more* concentrated than search demand over the same catalogue — a prediction the design makes and the literature does not, and the most testable claim in the paper.

Where the mechanism stops. It decides which stall you enter, not what you pick inside. Figure 4 shows why: the kiosk opens a product list, and at that point you are choosing from a list again. Labels, prices and familiarity bring search-like choice back within the stall. The claim is about how visits are spread across stalls, and about what a visitor is ever exposed to, not about choice inside a stall. Any test must measure at the stall level.

6.1 A line the design can cross without noticing

Finding by presence is a property of the format only as long as the format does not bring back the alternative. The prototype has an AI guide available at every site. A guide that answers questions about the temple is irrelevant to Proposition 1a. A guide that answers “where can I get a coffee” is not. At that moment discovery goes back to being a search, exposure goes back to depending on what you asked for, and position stops mattering.

This matters more than it seems, because the change is silent and commercially tempting. Recommendation is the first fix anyone would propose for a market with disappointing sales, and it would destroy the property that makes the format distinct while appearing to improve it. So we state

the condition plainly: *Proposition 1a holds only where product discovery is not mediated by search or recommendation. A market that adds conversational or algorithmic product suggestions is, for the purposes of this model, a searchable shop with spatial decoration.*

We treat this as a contribution rather than a caveat. It gives the format a testable boundary, and it identifies one specific design decision — whether the guide may recommend shops — as deciding which of two demand regimes a platform is in. You cannot tell from the interface, which is why it needs saying.

Proposition 2 (conjectural). *Supplying a product locally reduces how authentic buyers find it, and the reduction is larger the more the product’s value depends on being made where it comes from — what we call its origin-dependence.*

A coffee brewed locally after a virtual visit loses little. A textile whose whole value is having been woven in Cambodia loses everything. Origin-dependence is a construct we introduce here. Its nearest relatives are research on country-of-origin effects and the legal system of protected geographical names, neither of which treats it as a sliding scale a routing system could act on.

Proposition 3 (conjectural). *Local supply increases transaction value for low origin-dependence products and decreases it for high ones. Efficient routing therefore depends on origin-dependence as well as region, not on region alone.*

This is the practical claim. A system that routes purely by geography maximises convenience and destroys authenticity for exactly the products whose authenticity is their value. The implication is two lanes: local supply for fungible goods, supply from the origin for origin-dependent ones. Making origin-dependence a routing variable rather than a marketing label is, as far as we can find, new.

Proposition 4 (conjectural). *Where a locally supplied product carries a verifiable proof of origin, the loss of authenticity under Proposition 2 is partly offset, and the offset is larger the more independent the proof is from the seller.*

This proposition has a design precedent rather than an evidential one. A provenance system in which neither end holds cryptographic keys — developed for smallholder provenance in a design paper by one of us (Chay 2026a) — would let a producer at the origin prove a claim to a buyer who never travelled, without either party running key infrastructure. We cite it as a worked design argument and not as evidence; that paper states that it contains no empirical content.

6.2 The sample catalogue, and what it does and does not show

The sample products tell you nothing about what will be sold. The prototype’s goods are coffee, fried chicken and burgers, under invented names. They are placeholders chosen to test the routing with familiar categories and no real brand names. They are not a statement of what the market is for. We say this plainly because a reader could otherwise read a design intention into a test fixture, and the reading would be wrong.

What the placeholders do give us is a clean setting for one test. Their origin-dependence is zero. A burger has no link to a Khmer temple, and no buyer would think it did. That removes authenticity, provenance and origin value from the setting entirely — inconvenient for a market meant to carry Cambodian goods, convenient for an experiment. Propositions 2 to 4 are switched off there: no authenticity to lose, so local supply costs nothing and the two-lane rule trivially picks the local lane. Proposition 1b is untouched and stands alone. Any concentration effect measured in that setting is due to position and cannot be due to the goods.

That is a methodological convenience, not a description of the design. Origin-dependence and the two-lane rule are for later — for the point when real merchants and place-tied goods (textiles, spices, carving, craft) arrive, and the routing question stops being trivial. Nothing about the present catalogue tells us whether that guidance is right. It only tells us the system has not been asked the question yet.

7. Money: Charging for the Ground

7.1 The problem

Section 5.3 named the oddity; here is its consequence. The fulfiller has no contract with the platform: under geographic routing the merchant is chosen at checkout, so no contract can exist with the party whose demand was created. The platform does not take payment, by its own rule. Together these close off every way of earning from the sale.

Proposition 5 (analytic). *Where demand created in a place-anchored market is fulfilled by a provider with no contract with the platform, in the buyer’s own country, and the platform neither takes the payment nor charges a referral fee,*

the origin earns exactly nothing from the sale, however many sales there are.

This is what the two rules together imply. We are careful about what it does and does not show.

Creating demand that someone else fulfils, without a contract with them, is ordinary. Affiliate networks, search ads, map listings and social referrals all do it. What distinguishes them is that the originator *charges for the referral* — per click, per sale, per placement. This platform declines that too. So the unusual thing is not the missing contract with the fulfiller. It is that a platform generating commercially valuable traffic has not turned that traffic into money at any point in the transaction.

That is better described as a public-good platform choosing not to charge for its own referrals than as a failure of any theory. The theory of how firms enter foreign markets concerns a firm exploiting something it owns abroad and choosing a structure to keep the profits (Dunning 1988); it was not written for an entity that generates traffic and declines to price it. The nearer literature is Brouters and colleagues’ (2022) account of non-traditional market entry, which already includes digital entry with no local presence. The modest, defensible claim is that this case sits at the far end of that spectrum — no presence, no contract, no referral revenue — and that what remains to charge for is therefore not the sale but the ground.

7.2 The instrument

If the sale cannot be charged for, what remains is the ground. The platform controls one scarce thing: position in a market whose visitors its own content brings. That allows the familiar tools of market-hall management — rent for a stall, a fee for the right to trade, a premium for a spot near the busiest point, and anchor-tenant deals.

Proposition 6 (conjectural — a design proposal, not an entailment). *Where the platform declines to earn from the sale, a place-anchored platform could earn through location rent priced on the visitor flow its content generates, without breaking its rule against taking payment.*

The last clause is why this works rather than being merely a different price list. Rent is charged for space, not sales. The platform never touches the buyer’s money, never needs a payment licence, and never takes on cross-border settlement or identity-checking obligations. A landlord is not a payment processor. The rule against taking payment and the revenue model, which under a commission design pull against each other, stop pulling.

We label this conjectural on purpose. It does not follow from the design, because the design has no stall and no tenancy (Section 5.4). It is a proposal for an instrument that would need the rentable thing built first. A reader should treat it as a design argument the rails allow, not a property they show.

Proposition 7 (carried in part, conjectural in part). *Rent*

in a market entered by teleport rises with how much visitor attention a stall is expected to capture, not with its distance from a point of interest. And the scarcest position is the market's attachment to the site itself, not any spot within the market.

It is tempting to state this as a distance rule and cite the virtual land-value research as support. That would misread it. Goldberg, Kugler and Schär (2024) do find that location premiums survive in virtual worlds, but the reason is a spot's power to attract visitors, not travel cost, and their central result is a *teleportation threshold*: where people can teleport, distance stops shaping behaviour beyond a certain point. A smooth distance rule is close to the shape their result argues against.

That matters acutely here, because this market is entered by teleport from the site. Every visitor arrives at the same point. The relevant variable is position relative to the *arrival point*, not distance to the temple — a variable Goldberg and colleagues do not test, since their setting compares walking with teleporting across an open world. What their work does support is the attention channel: value goes to spots expected to capture visits. And it suggests that the scarce asset in this design is not the stall nearest the fountain but the right to be the market attached to the site at all.

7.3 Two limits

Proposition 8 (conjectural, with a documented precedent). *Rent is self-limiting. Above some level, merchants leave, the market empties, visitors engage less, and the flow the rent was priced on shrinks.*

We regard this as the most important qualification in the paper, because it is not hypothetical. Kuntze and colleagues name rising virtual land rent among the causes of the 2008 exodus. A platform proposing rent is proposing the mechanism that emptied the last generation of virtual markets. What this gives us is not a warning but an optimisation problem: rent is capped by its own effect on the asset it is priced against, and the cap is empirical.

Proposition 9 (analytic, if the tenant is a local merchant). *Where the platform's mission includes helping merchants from the origin country, a single rent for everyone works against that mission; better designs charge by where the merchant comes from.*

The tension is concrete. A platform justified partly as giving Cambodian small businesses a global shopfront, which then charges them rent for it, taxes the people it exists to help. This argues for two tiers — subsidised or free stalls for origin merchants, commercial rent for foreign brands — which is a cross-subsidy with plenty of precedent in physical market halls and none, as far as we can find, in platform economics.

Proposition 10 (analytic, on a documented constraint). *A virtual tenancy is a legal claim. Where the host coun-*

try lacks registered title with clear legal effect and a fair process for recovery, rent can be collected by agreement but not enforced in court, and the arrangement rests on the platform's power rather than on property rights.

A companion paper by several of us (Chay and colleagues 2026) identifies enforceability of claims as a binding constraint in Cambodia, documenting an unresolved defect in the ranking of claims and a low international score on that dimension. A virtual stall tenancy is exactly such a claim. We record the consequence and do not repair it.

We add one warning from the same companion paper that applies with force here. Putting land tenure on a ledger, in a country where land tenure has been contested, is not a neutral technical choice. The architecture has reserved fields that would allow stalls to be represented as tokens. We recommend against using them for tenancies, and recommend the instrument be a revocable licence to occupy, claiming no registered title — which is both more prudent and, given Proposition 10, a more honest description of what it can actually be.

This is a property of a country, not of the format. A market issued where registered title works would not face it. It belongs to the case discussion and does not travel with the definition in Section 5.5.

7.4 Measurement

Proposition 11a (analytic, on official definitions). *Because buyer and fulfiller are both resident in the buyer's country and the platform takes no payment, the goods sale is recorded as domestic commerce, adds nothing to the origin's exports, and falls outside the international framework for measuring digital trade.*

Proposition 11b (analytic). *The platform's own income is not covered by 11a. Rent charged to a foreign tenant, or attention sold as advertising, is a service supplied by a resident of the origin to a non-resident and can in principle be measured as trade in services.*

The international handbook on measuring digital trade (IMF, OECD, UNCTAD and WTO 2023) defines its object as trade that is digitally ordered or delivered, between residents of different countries. A purchase created by a Cambodian platform, fulfilled between two residents of the buyer's country, with no payment at the platform, meets neither test.

The gap is therefore specific to the goods sale. What is invisible is the commerce the market induces, which is the part a national platform would most want to point to. What remains visible is whatever the platform itself charges — which, under Proposition 6, is the only part it would ever receive. A national digital-economy programme whose induced commerce is invisible to the statistics that judge national digital-economy programmes is worth stating plainly.

8. Discussion

8.1 What the paper adds

To retail theory, the paper adds a way of finding things. The claim is not that 3D shops are more engaging — the atmosphere research owns that scale and does not need us. It is that finding by presence conditions what you see on position rather than on what you wanted, which is a structural property, not a matter of degree.

To the study of how firms enter foreign markets, the paper adds a limiting case. Brouthers and colleagues (2022) already include digital entry without local presence; this case goes further — no presence, no merchant contract, no referral revenue — and is best described as a public-good platform that generates valuable traffic and has chosen not to sell it at any point. That is not a theoretical failure. It is a choice, made twice by two design rules, whose consequence is that the only remaining instrument is the ground beneath the stall, and that instrument does not yet exist in the design.

To platform economics, the paper adds an instrument. Platform revenue is normally theorised as a share of the transactions the platform sits between. Where a platform's attraction is content it made, and its transactions are out of reach, the available instrument is location rent — the logic of a market hall, not of a two-sided marketplace.

8.2 Selling meaning without the thing

The uncomfortable implication of Propositions 2 to 4 deserves stating rather than burying, and it becomes live only once place-tied goods enter the market — which, as Section 6.2 notes, the sample catalogue does not yet do. A place-anchored market that supplies locally sells a place's meaning attached to a product made somewhere else. When the product is generic this is unremarkable. When it is not, the arrangement comes close to licensing a place's meaning to goods that have nothing to do with it, and the fact that the licensor is the origin community does not by itself settle whether that is legitimate.

We do not resolve this and we are not sure it resolves cleanly. Proposition 3's two-lane rule keeps origin-dependent goods on the origin lane, which addresses the economic form of the problem. It does not address the cultural form, which is about what a place is entitled to sell of itself. We note that the design's own land-use rule — commerce in market zones, never on sacred ground — is an attempt to answer a version of this question spatially, and a spatial answer may not be enough for a question that is not spatial.

8.3 Who gets the money

Sections 7.2 and 7.3 together produce an awkward result for a platform with a development mission. Earning from the sale is closed off. Earning from rent is open. But rent falls on merchants, and the merchants a development mission exists to help are the ones least able to pay. Proposition 9's two-tier design resolves this on paper. Whether it survives contact

with a rent base narrow enough to be worth collecting is an empirical question we cannot answer, and it is the first thing we would want to know.

8.4 What follows for platforms like this

Nothing here argues against building place-anchored commerce in an emerging economy. It argues for answering three questions first rather than later. Which lane does each product category belong on — and so what does the router need to know about goods that a geographic router does not. What kind of instrument is the tenancy, and is it enforceable where it will be issued. And how will the resulting activity be measured, given that the obvious statistics will not see it. Each can be answered in advance and is expensive to answer late.

9. What We Do Not Know

We would rather list the paper's limits than have them found.

There is no empirical content. The propositions marked conjectural are untested. Those marked analytic follow from stated premises and are not thereby true of the world, since a premise may fail. The platform's commerce layer has never processed a transaction, and nothing here is evidence about how such a system performs.

The case is single and we are connected to it. Section 3.5 describes the mitigations and we do not think they remove the concern.

The assumptions in Section 5.6 are assumed, not tested here. The presence assumption is best supported in the full-detail and VR modes and untested in the reduced mode. Having used the prototype on a cheap phone we can say the scene renders and can be navigated; we cannot say it produces presence in the sense the research measures, and the two should not be confused. We did not observe the higher modes, so we make no claim about them either. How the mechanism varies by mode is the most consequential thing this paper does not know.

Our observation of the prototype is itself limited: one point in time, one device, no instruments. Sections 4.5 to 4.7 separate what we saw from what is specified, but a reader should treat the figures as illustration, not measurement. In particular, a screen that shows a fulfilment country is not evidence that a fulfilment happened, and the fixed total noted in Section 4.6 means the commercial arrangement shown is not yet a coherent one.

Proposition 1b's concentration prediction runs against a well-established result in the other direction and rests on a mechanism we have reasoned about rather than observed.

We have not modelled competition. One market, one landlord, one captive attraction is the simplest case. What happens when several platforms anchor to virtual versions of the same real place is a question we have not asked.

Our reading of Su, Lai and Huang (2024) in Section 2.5 rests on their published abstract; we were unable to obtain the full text. The abstract establishes that a place-rendered

shop was not among their conditions, which is all the claim in Section 2.5 requires. It does not establish how their Internet condition was presented, so we make no claim about that. Their finding is empirical and ours is a hypothesis about a channel their study did not test.

10. What Should Happen Next

Four studies would move this materially. We state them in the order we would run them.

First, and it should be in the next version of this paper rather than promised after it: a study comparing the market against a plain product list over the same catalogue, testing Proposition 1b’s concentration claim at the stall level. It can be run now. The prototype’s plaza already places kiosks in distinct positions, so logging which stall each visitor enters, how long they stay and what they select needs no payments, no new construction and no merchants; rotating the kiosk positions between sessions separates the effect of position from the effect of the merchant. It tests the paper’s most testable claim at the lowest cost, and it would move this work from architecture plus reasoning to architecture plus one result.

Second, an experiment varying where a product is supplied from against what kind of product it is would test Propositions 2 and 3 directly, and would show whether origin-dependence behaves as a sliding scale or a threshold.

Third, a field test of the two routing lanes, once real fulfilment exists, would test Proposition 3 against conversion and satisfaction at once — the two may not move together, which would itself be the finding.

Fourth, a rent-elasticity study, on any operating virtual market, would locate the tipping point in Proposition 8. Given the 2008 record, this is the study whose absence matters most to anyone considering the model.

11. Conclusion

Virtual shops have been announced twice and abandoned twice, and the scepticism is earned. Our argument is not that the third attempt will succeed. It is that the first two never reached the point where the interesting questions start, because they built places to look and not places to buy — and a 3D shop that actually sells things has properties a showroom cannot have.

Those properties are unusual. The buyer arrives by encounter, so what she sees depends on where she stands rather than on what she wanted. The seller is chosen at checkout by geography, so no contract links the party who created the demand to the party who fills the order. The platform takes no payment and charges no referral fee, so the sale is out of reach at the core. And what could be charged for, if anything, is the ground — which would keep the no-payment rule intact, would inherit the failure that emptied the last generation, and in the country examined would rest on a claim that can be collected but not enforced.

The chain runs one way, and it stops short of where an

earlier reading of it ended. Because the shop is a place, the buyer encounters it. Because she encounters it, the seller is not yet known. Because the seller is not known and the core will not take payment, the sale is out of reach. Nothing in the design then entails rent. The rails define a router and a list of providers, not a lease. If the platform is to earn anything, it will first have to invent a rentable thing that its own specification does not yet contain — and decide who the tenant is. That is a coherent architecture up to the point where it stops, and it is not the one anybody set out to build.

Declarations

Conflict of interest. The authors declare no commercial or financial relationships that could be construed as a potential conflict of interest. All authors hold appointments at the National University of Management. The first author directs the CamboVerse Center, an institute at that University; the third and fourth authors are engineers on the CamboVerse project. These are affiliations. No institutional relationship between the CamboVerse project and the Center, the University, or any external body is asserted anywhere in this paper.

Some authors hold senior institutional roles, so this bears emphasis: participation here is as individual researchers, the paper is the authors’ own academic work rather than institutional output, it was not produced through, reviewed by, or approved under any University governance process, it is not a University position, policy or endorsement, and it should not be cited as one.

The authors have an evident interest in the case appearing significant; the mitigating circumstance is that the case record is public and every descriptive claim in Section 4 is independently checkable.

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Author contributions. SC conceived the study, developed the constructs and propositions, and drafted the manuscript. RP contributed the institutional and economic analysis and the policy argument of Section 8. LK and MV designed and implemented the commerce architecture and prototype described in Section 4 and verified the architectural description reported there. All authors reviewed and approved the submitted version and accept responsibility for its content.

Relationship to prior work. This paper draws its institutional constraints from a companion paper by several of the present authors on digital economy transition, and its provenance mechanism in Proposition 4 from a design paper by the first author on smallholder provenance infrastructure. Neither is a source of empirical evidence for the claims made here: the provenance paper states that it contains no empirical content, and the companion paper’s evidence concerns a ledger-layer deployment rather than any commercial activity. The present paper’s contribution — the discovery-topology argument, the three-role decomposition of a settlement-abstaining precinct, and the ground-rent instrument as a design proposal — is developed here for the first time and does not depend on either.

Data availability. No data were generated or analysed. The architecture described in Section 4 is published in full in the CamboVerse public repository (<https://github.com/camboversecenter/CamboVerse>), and the prototype recorded in Figures 1 to 5 is publicly reachable at preview.camboverse.world. Every descriptive claim in Section 4 can be checked against one or both. Interface states are reported as observed on 4 September 2026; the prototype is under active development and will change.

Generative AI disclosure. The authors used Anthropic’s Claude (Opus 5) during the preparation of this manuscript for prior-art search across the literatures named in Section 3.3, verification of reference metadata against publisher records, structural editing, drafting and revision of text, and preparation of the typeset source. The paper’s central claims originated with the authors: that what changes in this configuration is the retail form rather than the trade structure, and that ground rent is the residual instrument — as a design proposal — for a platform that abstains from settlement. Prior-art search materially narrowed the contribution claimed, and several positions advanced in earlier drafts were withdrawn as a result; Section 3.3 records this. The authors reviewed all AI-assisted output,

verified the cited sources, and take full responsibility for the content of this paper.

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